Zhang's Method for Planer-based Camera Calibration.

→ ① Images $I_0 \cdots I_{M-1}$ (total M images) of the Model are taken under diff. views by moving it around

→ ② for each image I_i ($i = 0, \ldots M-1$) N-sensor points $u_{i,0} \cdots u_{i,j} \cdots u_{i,N-1}$ are extracted

$\Rightarrow u_{i,j}$ $\quad j = (j = 0 \rightarrow N-1)$

$i = (0 \rightarrow M-1)$

are observed in 1:1 correspondence

$u_{00}, u_{01}, u_{02} \ldots u_{0N-1}$

$u_{10}, u_{11} \ldots u_{1,N-1}$

$\vdots$

$u_{M-1,0} \quad u_{M-1,N-1}$.

Where $u_{i,j}$ is one pair-co-ordinate of ith image.

$u_{i,j} = (x_i, y_i)$

7x7 chessboard → 49 pts.

$\Rightarrow u_{00} \ldots u_{0,49}$

$u_{10} \ldots u_{1,49}$

$u_{20} \ldots u_{2,49}$ & so on.

→ ③ From observed points (for each layout/pose of chessboard / each image) the associated Homographs

$H_0 \ldots \rightarrow \ldots H_{M-1}$

are estimated.

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→ (4) From homography H_i compute 5 intrinsic params (α, β, γ, u_c, v_c) of camera using closed form (linear) solution.

→ (5) Once intrinsics are known, compute ($\vec{R_i}$, $\vec{t_i}$) Rotation & translation vectors w.r.t. 3D extrinsic parameters.

→ (6) Compute k_0, k_1 by linear least squares. (Radial distortion parameters)

→ (7) Use estimated initial values as initial guess for optimizing over H-all-view using levenberg marquardt (lm) optimizer.

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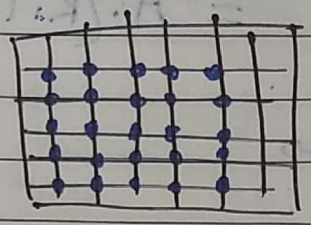
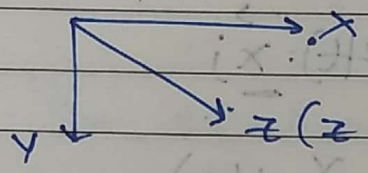
Computation of Homography.



Checkboard pattern
$= 5 \times 5$



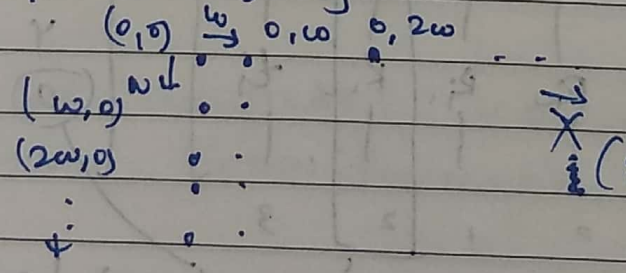
z (z is normal to page)
Checkboard
Hence for All 3D pts,
$\boxed{z = 0}$

$\therefore$ N reference pts : M model images.
$\hookrightarrow$ $x_0 \ldots x_{N-1}$ whose 3D coordinates are known

$u_{i,j} \in \mathbb{R}^2$

$\therefore u_{ij} \longleftrightarrow x_{ij}$

Hence compute x_{ij} by:



Moving on to homography estimation.

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Homography estimation for each view.

Let image coordinate be $(u_{i,j}, v_{i,j})$

$$\therefore \quad s \cdot \begin{pmatrix} u_{i,j} \\ v_{i,j} \\ 1 \end{pmatrix} = \underbrace{A}_{\text{Intrinsic}} \cdot \underbrace{(R_i | t_i)}_{\text{Extrinsic}} \cdot \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

homogeneous Image co.

Homogeneous world co

$$\therefore \quad s \cdot \vec{u}_{ij} = A \cdot (R|t) \cdot \vec{x}_j$$

where $A = \begin{pmatrix} \alpha & \gamma & u_c \\ 0 & \beta & v_c \\ 0 & 0 & 1 \end{pmatrix}$

Now $z = 0 \quad \forall z \in X_j$

$$\Rightarrow s \begin{pmatrix} u_{i,j} \\ v_{i,j} \\ 1 \end{pmatrix} = \underbrace{A}_{3\times3} \cdot \underbrace{(R_i | t_i)}_{3\times4 \text{ matrix}} \begin{pmatrix} X \\ Y \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{array}{c} 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} | & | & \boxed{\phi} & | \\ R_{i0} & R_{i1} & \boxed{R_{i2}} & t_i \\ | & | & \boxed{|} & | \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{array}{cccc} 0 & 1 & 2 & 3 \end{array}$$

leads to zero

$$\Rightarrow s \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = A \cdot (R_{i0}, R_{i1}, t_i) \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$\therefore s \begin{pmatrix} u_{ij} \\ v_{ij} \\ 1 \end{pmatrix} = A \begin{pmatrix} R_{i0}, R_{i1} & t_i \end{pmatrix} \begin{bmatrix} X_{ij} \\ Y_{ij} \\ 1 \end{bmatrix}$$

reduces one dimensi

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$$\Rightarrow s \begin{pmatrix} u_{ij} \\ v_{ij} \\ 1 \end{pmatrix} = \underbrace{H_i}_{\substack{\text{Homography} \\ \text{for view}_i}} \begin{pmatrix} x_j \\ y_j \\ 1 \end{pmatrix}$$

$$\Rightarrow H_i \Rightarrow \begin{pmatrix} | & | & | \\ h_{i0} & h_{i1} & h_{i2} \\ | & | & | \end{pmatrix}$$

Which is equivalent to $\lambda \cdot A \cdot (r_{i0}, r_{i1}, t_i)$

$(\lambda = 1/s)$

$\therefore$ Now the task boils down to

$$\begin{pmatrix} u_{ij} \\ v_{ij} \\ 1 \end{pmatrix} \xrightarrow{H_i} \begin{pmatrix} x_j \\ y_j \\ 1 \end{pmatrix}$$

or $(u_{ij}, v_{ij})^T \xrightarrow[H_i]{} (x_j, y_j)^T$

This can be computed using DLT or direct linear transformation

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Using DLT for Homography Estimation

Let $\mathcal{X} = (x_0, \dots x_{N-1})$ with $x_j = (x_j, y_j)$
and $\mathcal{U} = (u_0, \dots u_{N-1})$ with $u_j = (u_j, v_j)$

$\therefore \underline{u_j} = H \cdot \underline{x_j}$

or
$$\begin{pmatrix} u_j \\ v_j \\ w_j \end{pmatrix} = \begin{pmatrix} H_{00} & H_{01} & H_{02} \\ H_{10} & H_{11} & H_{12} \\ H_{20} & H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} x_j \\ y_j \\ z_j \end{pmatrix} \quad z_j \to 1$$

$$\Rightarrow \begin{pmatrix} u_j \\ v_j \\ w_j \end{pmatrix} = \begin{pmatrix} H_{00} & H_{01} & H_{02} \\ H_{10} & H_{11} & H_{12} \\ H_{20} & H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} x_j \\ y_j \\ 1 \end{pmatrix}$$

* $$u_j = \frac{u_j}{w_j} = \frac{H_{00} \cdot x_j + H_{01} \cdot y_j + H_{02}}{H_{20} x_j + H_{21} \cdot y_j + H_{22}}$$

* $$v_j = \frac{v_j}{w_j} = \frac{H_{10} \cdot x_j + H_{11} \cdot y_j + H_{12}}{H_{20} x_j + H_{21} \cdot y_j + H_{22}}$$

$$\Rightarrow u_j (H_{20} x_j + H_{21} y_j + H_{22}) = H_{00} x_j + H_{01} y_j + H_{02}$$
$$v_j (H_{20} x_j + H_{21} y_j + H_{22}) = H_{10} x_j + H_{11} y_j + H_{12}$$

$\therefore$ ~~$u_j x_j H_{20}$ $+ u_j y_j H_{21} + u_j$~~

$$\begin{pmatrix} -x_j & -y_j & -1 & 0 & 0 & 0 & u_j x_j & u_j y_j & u_j \\ 0 & 0 & 0 & -x_j & -y_j & -1 & v_j x_j & v_j y_j & v_j \end{pmatrix} h = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

where $h = (H_{00}, H_{01}, H_{02}, H_{10}, H_{11}, H_{12}, H_{20}, H_{21}, H_{22})^T$

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N points in one image

Hence for ~~m~~ - images:

$$
\begin{array}{c}
\text{Im 0}\left\{\begin{array}{c}
\begin{pmatrix} -x_0 & -y_0 & -1 & \cdots \\ 0 & 0 & 0 & \cdots \end{pmatrix} \\
\left\{\begin{pmatrix} -x_1 & -y_1 & -1 \\ 0 & 0 & 0 \end{pmatrix}\right. \\
\left\{\begin{pmatrix} -x_2 & -y_2 & -1 \\ \vdots & \vdots & \vdots \end{pmatrix}\right. \\
\vdots \\
\begin{pmatrix} -x_{N-1} & -y_{N-1} & -1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & -x_{N-1} & -y_{N-1} & -1 & \cdots \end{pmatrix}
\end{array}\right.
\end{array}
\vec{h} = \begin{array}{c} 0 \\ \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \\ N-1 \end{array}
$$

$$\underset{\text{(matrix)}}{\vec{M}} \cdot \vec{h} = 0$$

Structure of $\vec{M}$

$$
\begin{array}{c}
\left\{\begin{array}{c}
\begin{pmatrix} -x_0 & -y_0 & -1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & -x_0 & -y_0 & -1 & \cdots \end{pmatrix} \\
\left\{\begin{pmatrix} \\ \\ \end{pmatrix}\right. \\
\vdots \\
\left\{\begin{pmatrix} \\ \\ \end{pmatrix}\right.
\end{array}\right. \\
\longleftrightarrow 9
\end{array}
$$

x_0, x_1, x_n

2N rows. $\therefore \vec{M}$ is order of $(2N \times 9)$

$\vec{M} \rightarrow (2N \times 9)$ elements per image.

$$M = U \cdot S \cdot V^T \qquad \because SVD(M)$$

U: $2N \times 2N$, S: $2N \times 9$, V^T: 9×9 unitary mtx.

$$M \underset{SVD}{\Rightarrow} \begin{pmatrix} u_{00} & \cdots & u_{0,2N-1} \\ u_{10} & \cdots & u_{1,2N-1} \\ \vdots & & \vdots \\ u_{2N-1,0} & & u_{2N-1,2N-1} \end{pmatrix} \begin{pmatrix} s_0 & & & & \\ & s_1 & & & \\ & & s_2 & & \\ & & & \ddots & \\ & & & & s_8 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_{00} & & v_{80} \\ v_{09} & \cdots & v_{81} \\ \vdots & & \vdots \\ v_{08} & & v_{88} \end{pmatrix}$$

cause its transpose.

$$\therefore M = \begin{pmatrix} | & | & | & & | \\ u_0 & u_1 & u_2 & \cdots & u_{2N-1} \\ | & | & | & & | \end{pmatrix} \cdot S \cdot \begin{pmatrix} - & v_0^T & - \\ - & v_1^T & - \\ - & \vdots & - \\ - & v_8^T & - \end{pmatrix}$$

Let $S_k = \min(s_0, \ldots s_8)$

k = row index of min. eigen value

$$\boxed{\vec{h} = v_k \qquad \text{with } k = \underset{0 \le i < 9}{\operatorname{argmin}} S_i}$$

→ For computing $\vec{h}$ using DLT, center data / Normalize

$\therefore$ Let $x' =$ Normalize $(x) \Rightarrow x' = \vec{N_x} \cdot \vec{x}$

$u' =$ Normalize (u) $\quad u' = \vec{N_u} \cdot \vec{u}$

then compute $H' =$ Homography for each packet of N-points in each image.

finaley $H = N_u^{-1} \cdot H' \cdot N_x$.

Hence.

Input : correspondances $(\vec{x}, \vec{u})$

$x' = \vec{N_x} \cdot \vec{x}$

$u' = \vec{N_u} \cdot \vec{u}$

$H_i' =$ Homography (x_i', u_i')

$H_i =$ Denormalize (H_i')

$\hookrightarrow H = N_u^{-1} \cdot H' \cdot N_x$

$\therefore$ For each image i,

Compute H_i

$\therefore H_i =$ Set of Homographies $i \in [0, M-1]$

Hence goal is to ~~minimize~~ find optimum homography to minimize reprojection error.

$$E_{proj} = \sum_{j=0}^{N-1} \| u_j - H \cdot x_j \|^2$$

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Computing $\vec{N_x}$

$$N_x = \begin{pmatrix} S_x & 0 & -S_x\,\bar{x} \\ 0 & S_y & -S_y\,\bar{y} \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore N_x^{-1} = \begin{pmatrix} \frac{1}{S_x} & 0 & \bar{x} \\ 0 & \frac{1}{S_y} & \bar{y} \\ 0 & 0 & 1 \end{pmatrix}$$

$$(\bar{x}, \bar{y})^T = \frac{1}{N} \sum_{j=0}^{N-1} x$$

$$\sigma_x^2 = \frac{1}{N} \sum_{j=0}^{N-1} (x_j - \bar{x})^2 \qquad \sigma_y^2 = \frac{1}{N} \sum_{j=0}^{N-1} (y_j - \bar{y})^2$$

$$\therefore S_x = \sqrt{\frac{2}{\sigma_x^2}} \qquad S_y = \sqrt{\frac{2}{\sigma_y^2}}$$

$$N_x = \begin{pmatrix} S_x & 0 & -S_x \cdot \bar{x} \\ 0 & S_y & -S_y \cdot \bar{y} \\ 0 & 0 & 1 \end{pmatrix}$$

$\longrightarrow$ return normalization matrix.

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Zhang's Paper
M-TR

→ Let A = initial intrinsic matrix.

$$\therefore \overrightarrow{p(u,v)} = A \cdot [r \mid t] \cdot \overrightarrow{p(x,y,z)}$$

$$\Rightarrow s\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A \cdot [r_1 \; r_2 \; t] \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (\because z=0)$$

Let $H = A[r_1, r_2 \; t]$.

$$\therefore s\tilde{m} = H\tilde{M}$$

↳ H = view homography (3×3)

H computed by $M \cdot \vec{h} = 0$ with $|\vec{h}| \to 1$
minimization problem (Levenberg Marquardt)

$$\therefore \text{Let } H = \begin{bmatrix} | & | & | \\ h_1 & h_2 & h_3 \\ | & | & | \end{bmatrix}$$

$$\therefore [h_1 \; h_2 \; h_3] = \lambda A [r_1 \; r_2 \; t]$$

Now r_1, r_2 are orthonormal

$$\Rightarrow \vec{r_1} \perp \vec{r_2} \;\&\; \|r_1\| = \|r_2\| = 1$$

$\Rightarrow h_1 = \lambda \cdot A \cdot r_1$ | $r_0^T \cdot r_1^T = r_1^T \cdot r_0 = 0$
$h_2 = \lambda \cdot A \cdot r_2$ | (All dot product = 0)

$$\Rightarrow h_1^T A^{-T} \cdot A^{-1} \cdot h_2 = 0$$

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Computing A

$$B = (A^{-1})^T \cdot A^{-1}$$

$$u_1^T \cdot (A^{-1})^T \cdot A^{-1} \cdot u_2 = 0$$

$$B = \begin{pmatrix} B_0 & B_1 & B_3 \\ B_1 & B_2 & B_4 \\ B_3 & B_4 & B_5 \end{pmatrix}$$

or

$$\begin{pmatrix} B_{11} & B_{24} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{pmatrix}$$

$$v_0 = (B_1 \cdot B_3 - B_0 B_4) / (B_0 \cdot B_2 - B_1^2)$$

$$\lambda = \{B_5 - [B_3^2 + v_0 (B_1 \cdot B_2 - B_0 B_4)]\} / B_0$$

$$\alpha = \sqrt{\lambda / B_0}$$

$$\beta = \sqrt{\lambda \cdot B_0 / (B_0 B_2 - B_1^2)}$$

$$\gamma = (-1) \cdot (v_0) \cdot \alpha^2 \times (\beta / \lambda)$$

$$u_0 = u_c = \gamma \cdot v_0 / \beta - B_3 \cdot \alpha^2 / \lambda$$

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